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MLDL Experiment 04

Aim: Implement K-Nearest Neighbors (KNN) and evaluate model performance

1. Dataset Source:

- **Dataset Name:** Breast Cancer Wisconsin Dataset
- **Source:** https://www.kaggle.com/datasets/uciml/iris
- **Original Source:** UCI Machine Learning Repository (Dr. William H. Wolberg)

2. Dataset Description:

The Breast Cancer Wisconsin dataset is used for **binary classification** of breast tumors as **benign** or **malignant**. The dataset contains features computed from digitized images of a fine needle aspirate (FNA) of a breast mass. These features describe characteristics of the cell nuclei present in the image.

Dataset Size

- **569 rows × 33 columns**

Target Variable

- **diagnosis**

Class Labels

- **M = Malignant**
- **B = Benign**

Preprocessing Note

Before training the model:

- The `id` column was removed because it does not contribute to prediction.
- The `Unnamed: 32` column was removed because it contains no useful information.
- The target variable `diagnosis` was converted into binary form:
  - **Malignant = 1**
  - **Benign = 0**

Features:-

The dataset contains numeric features such as:

1. radius_mean
2. texture_mean
3. perimeter_mean
4. area_mean
5. smoothness_mean
6. compactness_mean
7. concavity_mean

8. concave points_mean

9. symmetry_mean

10. fractal_dimension_mean

It also includes standard error and worst-value versions of these measurements.

These features are useful in identifying whether a tumor is benign or malignant.

3. Mathematical Formulation of the Algorithm:

KNN is a non-parametric, distance-based algorithm.

A. Distance Metric (Euclidean Distance):

$$d(x, x_i) = \sqrt{\sum_{j=1}^{n} (x_j - x_{ij})^2}$$

Where:

- n = number of features (4)
- x = new observation
- x_i = training sample

B. Classification Rule:

1. Compute distance between new point and all training samples

2. Select K nearest neighbors

3. Assign class using majority voting

4. Algorithm Limitations:

- Computationally expensive for large datasets
- Sensitive to feature scaling
- Performance depends on optimal choice of K

5. Methodology / Workflow:

1. Upload dataset
2. Drop Id column
3. Separate features and target
4. Apply StandardScaler
5. Split dataset (80% train, 20% test)
6. Train KNN model (K=5)
7. Evaluate using accuracy and confusion matrix
8. Tune K using Elbow Method

6. Performance Analysis:

Phase 1: Baseline KNN Model

A baseline KNN classifier was trained using:

- **n_neighbors = 5**

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Observation:

The Breast Cancer dataset is highly suitable for KNN because it contains multiple numeric features that describe tumor characteristics. Since KNN is distance-based, **feature scaling is mandatory** to ensure that all features contribute fairly to the distance calculation.

Phase 2: Error Rate Analysis

To determine a suitable value of K, the error rate was calculated for K values from **1 to 30**.

Observation:

The **Error Rate vs K Value** graph helps identify the range of K values where the classification error is minimum. This supports selecting an optimal K value for improved prediction performance.

Phase 3: Hyperparameter Tuning

Hyperparameter tuning was performed using **GridSearchCV** to identify the best combination of:

- `n_neighbors`
- `weights`
- `metric`

Overall Observation:

The tuned KNN model performed better than the baseline model because the optimal choice of hyperparameters improved classification accuracy and model generalization.

7. Hyperparameter Tuning:

The main hyperparameter in KNN is the number of neighbors (**K**). Selecting an appropriate value of K is crucial for achieving optimal performance.

- Small K values (e.g., K = 1) may lead to **overfitting**, making the model sensitive to noise.
- Large K values may lead to **underfitting**, producing overly smooth decision boundaries.

To determine the optimal K, the **Elbow Method** was applied by computing the error rate for K values ranging from 1 to 20 and plotting the error curve.

- The value of K corresponding to the **minimum error rate** was selected as the optimal K.
- After tuning, the model achieved stable and high accuracy, confirming improved generalization performance.

Hyperparameter tuning helped in selecting a balanced K value that minimizes classification error and enhances the robustness of the KNN model.

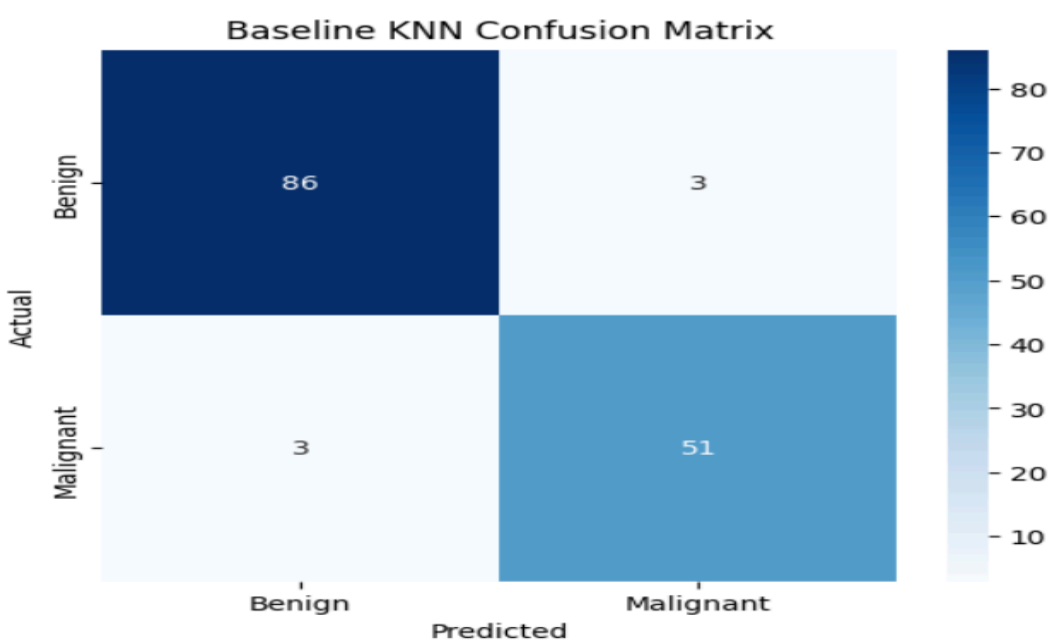
Code And Output:

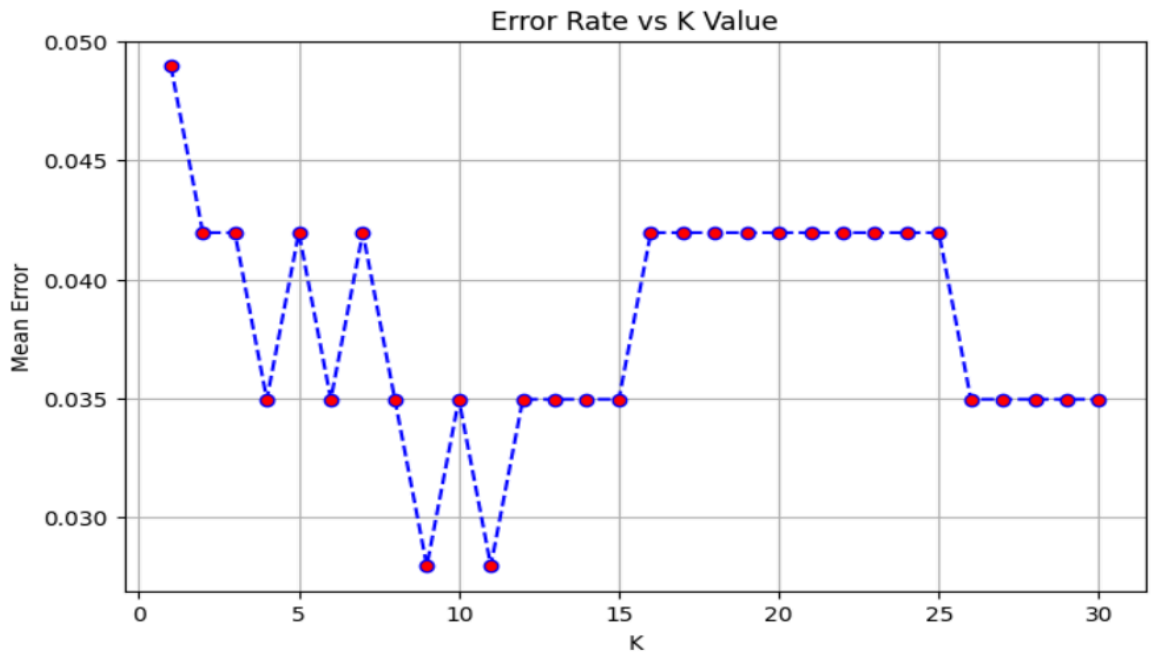
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection
import train_test_split,
GridSearchCV
from sklearn.preprocessing import
StandardScaler
from sklearn.neighbors import
KNeighborsClassifier
from sklearn.metrics import
accuracy_score, precision_score,
recall_score, f1_score,
confusion_matrix

from google.colab import files
uploaded = files.upload()

file_name =
list(uploaded.keys())[0]


df = pd.read_csv(file_name)

print("===== Dataset Preview
=====")
print(df.head())

df = df.drop(['id', 'Unnamed:
32'], axis=1)

df['diagnosis'] =
df['diagnosis'].map({'M':1,
'B':0})

print("\nMissing Values Check:")
print(df.isnull().sum())

X = df.drop('diagnosis',
axis=1).values
y = df['diagnosis'].values

X_train, X_test, y_train, y_test =
train_test_split(
    X, y, test_size=0.25,
random_state=42
)

scaler = StandardScaler()
X_train_scaled =
scaler.fit_transform(X_train)
X_test_scaled =
scaler.transform(X_test)

baseline_knn =
KNeighborsClassifier(n_neighbors=5
)
baseline_knn.fit(X_train_scaled,
y_train)

y_pred_baseline =
baseline_knn.predict(X_test_scaled
)

baseline_acc =
accuracy_score(y_test,
y_pred_baseline) * 100
baseline_prec =
precision_score(y_test,
y_pred_baseline) * 100
baseline_rec =
```

```
recall_score(y_test,
y_pred_baseline) * 100
baseline_f1 = f1_score(y_test,
y_pred_baseline) * 100

print("\n--- BASELINE MATHEMATICAL
RESULTS (k=5) ---")
print(f"Accuracy :
{baseline_acc:.2f}%")
print(f"Precision:
{baseline_prec:.2f}%")
print(f"Recall   :
{baseline_rec:.2f}%")
print(f"F1 Score :
{baseline_f1:.2f}%")

plt.figure(figsize=(6,5))
cm = confusion_matrix(y_test,
y_pred_baseline)
sns.heatmap(cm, annot=True,
fmt='d', cmap='Blues',

xticklabels=['Benign','Malignant']
,

yticklabels=['Benign','Malignant']
)
plt.title("Baseline KNN Confusion
Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

error_rate = []

for i in range(1,31):
    knn =
KNeighborsClassifier(n_neighbors=i
)
    knn.fit(X_train_scaled,
y_train)
    pred_i =
knn.predict(X_test_scaled)

error_rate.append(np.mean(pred_i
!= y_test))

plt.figure(figsize=(8,5))
plt.plot(range(1,31), error_rate,
         color='blue',
linestyle='dashed',
         marker='o',
markerfacecolor='red')
plt.title("Error Rate vs K Value")
plt.xlabel("K")
plt.ylabel("Mean Error")
plt.grid(True)
plt.show()

param_grid = {
    'n_neighbors':
np.arange(1,31),
    'weights':
['uniform','distance'],
    'metric':
['euclidean','manhattan']
}

grid_search = GridSearchCV(
    KNeighborsClassifier(),
    param_grid,
    cv=5,
    scoring='accuracy',
    n_jobs=-1
)

grid_search.fit(X_train_scaled,
y_train)

best_knn =
grid_search.best_estimator_
y_pred_tuned =
```

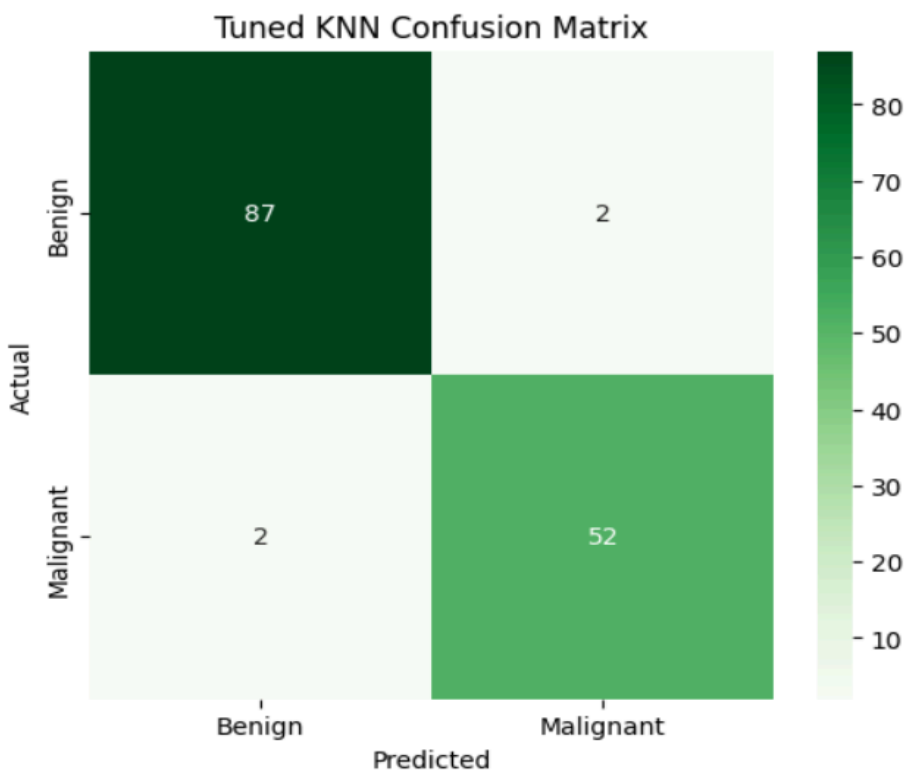
```
best_knn.predict(X_test_scaled)

tuned_acc = accuracy_score(y_test,
y_pred_tuned) * 100
tuned_prec =
precision_score(y_test,
y_pred_tuned) * 100
tuned_rec = recall_score(y_test,
y_pred_tuned) * 100
tuned_f1 = f1_score(y_test,
y_pred_tuned) * 100

print("\n--- TUNED MATHEMATICAL
RESULTS ---")
print("Best Parameters:",
grid_search.best_params_)
print(f"Accuracy :
{tuned_acc:.2f}%")
print(f"Precision:
{tuned_prec:.2f}%")
print(f"Recall   :
{tuned_rec:.2f}%")
print(f"F1 Score :
{tuned_f1:.2f}%")

plt.figure(figsize=(6,5))
cm_tuned =
confusion_matrix(y_test,
y_pred_tuned)
sns.heatmap(cm_tuned, annot=True,
fmt='d', cmap='Greens',

xticklabels=['Benign','Malignant']
,
yticklabels=['Benign','Malignant']
)
plt.title("Tuned KNN Confusion
Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

results_df =
pd.DataFrame(grid_search.cv_result
s_)

best_metric =
grid_search.best_params_['metric']
subset =
results_df[results_df['param_metri
c'] == best_metric]

pivot_table = subset.pivot(
    index='param_n_neighbors',
    columns='param_weights',
    values='mean_test_score'
)

plt.figure(figsize=(8,10))
sns.heatmap(pivot_table,
cmap='RdYlGn')
plt.title(f"Hyperparameter Tuning
Heatmap (Metric={best_metric})")
plt.ylabel("K Neighbors")
plt.xlabel("Weight Strategy")
plt.show()
```

```
===== Dataset Preview =====
         id diagnosis  radius_mean  texture_mean  perimeter_mean  area_mean  \
0    842302         M        17.99         10.38          122.80     1001.0
1    842517         M        20.57         17.77          132.90     1326.0
2  84300903         M        19.69         21.25          130.00     1203.0
3  84348301         M        11.42         20.38           77.58      386.1
4  84358402         M        20.29         14.34          135.10     1297.0

   smoothness_mean  compactness_mean  concavity_mean  concave points_mean  \
0          0.11840           0.27760          0.3001              0.14710
1          0.08474           0.07864          0.0869              0.07017
2          0.10960           0.15990          0.1974              0.12790
3          0.14250           0.28390          0.2414              0.10520
4          0.10030           0.13280          0.1980              0.10430

   ...  texture_worst  perimeter_worst  area_worst  smoothness_worst  \
0  ...          17.33           184.60      2019.0            0.1622
1  ...          23.41           158.80      1956.0            0.1238
2  ...          25.53           152.50      1709.0            0.1444
3  ...          26.50            98.87       567.7            0.2098
4  ...          16.67           152.20      1575.0            0.1374

   compactness_worst  concavity_worst  concave points_worst  symmetry_worst  \
0             0.6656           0.7119                0.2654          0.4601
1             0.1866           0.2416                0.1860          0.2750
2             0.4245           0.4504                0.2430          0.3613
3             0.8663           0.6869                0.2575          0.6638
4             0.2050           0.4000                0.1625          0.2364

   fractal_dimension_worst  Unnamed: 32
0                  0.11890          NaN
1                  0.08902          NaN
2                  0.08758          NaN
3                  0.17300          NaN
4                  0.07678          NaN

[5 rows x 33 columns]


--- BASELINE MATHEMATICAL RESULTS (k=5) ---
Accuracy : 95.80%
Precision: 94.44%
Recall   : 94.44%
F1 Score : 94.44%
```






```
--- TUNED MATHEMATICAL RESULTS ---
Best Parameters: {'metric': 'manhattan', 'n_neighbors': np.int64(3), 'weights': 'uniform'}
Accuracy : 97.20%
Precision: 96.30%
Recall   : 96.30%
F1 Score : 96.30%
```



Interpretation:

The confusion matrix shows how well the KNN model classifies breast tumor samples into benign and malignant categories.

- The rows represent actual classes.
- The columns represent predicted classes.
- Diagonal values represent correct predictions.
- Off-diagonal values represent classification errors.

In the **baseline model (K=5)**, most predictions lie on the diagonal, indicating that the classifier correctly identifies the majority of tumor cases. A few misclassifications are observed where benign tumors are predicted as malignant and vice versa.

After **hyperparameter tuning**, the confusion matrix shows improved performance. The number of correct predictions increases (87 benign and 52 malignant), while misclassifications decrease (only 2 in each case). This indicates better classification accuracy and improved model reliability.

Overall, the tuned KNN model demonstrates higher accuracy and better generalization compared to the baseline model.

This graph shows how the error rate changes as the number of neighbors (K) increases.

- The X-axis represents different values of K.
- The Y-axis represents the classification error rate.

When K is very small, the model is sensitive to noise and may overfit the training data, resulting in higher error.

As K increases, the error rate initially decreases, indicating improved model performance.

The minimum error is observed around **K = 9 to 11**, which represents the optimal range for K (Elbow Point).

After this point, the error stabilizes and slightly increases, suggesting that very large K values may lead to underfitting.

Selecting the optimal K improves model stability, reduces error, and ensures better prediction performance on unseen data.

Conclusion:

In this experiment, the **K-Nearest Neighbors (KNN)** algorithm was implemented on the **Breast Cancer Wisconsin Dataset** for binary classification. The goal was to classify tumors as **benign** or **malignant** using multiple numeric features extracted from breast mass cell nuclei.

The dataset was preprocessed by removing unnecessary columns such as `id` and `Unnamed: 32`, and the target variable `diagnosis` was converted into binary form where **Malignant = 1** and **Benign = 0**. Feature scaling was performed using **StandardScaler**, which is essential for KNN because it depends on distance calculations.